

P03. Cooperation Method of Connected and Automated Vehicles at Unsignalized Intersections: Lane Changing and Arrival Scheduling

中英双语博士精读笔记

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Theme / 主题: Lane changing + arrival scheduling

Method family / 方法族: Lane assignment + arrival scheduling / 车道分配与到达调度

PDF pages / 页数: 14 **Relevance / 相关度:** 82/100

Source / 来源: <https://arxiv.org/abs/2109.14175>

1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于“Lane changing + arrival scheduling”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Lane changing + arrival scheduling. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点: arXiv note reports text overlap with the authors' earlier graph-based paper; use mainly for lane-changing extension context.

1.2 核心科学问题 / Core Scientific Question

当车辆可以变道或选择目标车道时，如何联合决定路径/车道与无冲突到达时间？

How should target-lane assignment and conflict-free arrival scheduling be coupled when vehicles may change lanes before the intersection?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Two-stage framework decoupling lateral lane assignment/path planning from graph-based arrival scheduling with minimum clique cover.

- 安全/避碰贡献 (safety contribution): Conflict-free arrival scheduling and formation-control assumptions prevent incompatible movements.
- 平滑/停止避免贡献 (smoothing contribution): Improves traffic efficiency by choosing target lanes and arrival schedules that avoid unnecessary queuing.
- 创新力度评判 (reviewer judgement): 中等: 在 P02 图排序基础上加入横向决策, 扩展性有价值, 但理论新颖性主要来自组合而非新算法。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Lane assignment + arrival scheduling	车道分配与到达调度	该论文的核心方法族。

2 教材式学习路径与关键截图 / Textbook-Style Study Path & PDF Snapshots

2.1 学习路径 / Study Path

- 第一遍先读首页截图、摘要与引言, 确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading, 对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验, 判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

2.2 博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式: 联合车道分配与到达时间优化 / Joint lane assignment and arrival scheduling, 并解释每个符号的物理意义?
- 能否画出输入-模型-算法-控制-验证的数据流图?
- 能否指出至少两个强假设, 并说明它们在真实交通中何时失效?
- 能否复述实验基准是否公平, 指标是否覆盖效率、安全、能耗和计算时间?
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验?

2.3 关键截图讲解 / PDF Snapshot Explanation

Screenshots are selected anchors from the local PDFs for study and parallel reading. They do not replace the original paper.

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Cooperation Method of Connected and Automated Vehicles at Unsignalized Intersections: Lane Changing and Arrival Scheduling

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Abstract—The cooperation of connected and automated vehicles (CAVs) has shown great potential in improving traffic efficiency during intersection management. Existing research mainly focuses on intersections where lane changing is prohibited, which is impractical for real-life implementation. This paper proposes a two-stage cooperation framework, which decouples the longitudinal and lateral control of CAVs, allowing them to change to their preferred lanes. Based on formation control, an iterative framework is initially proposed to solve the target assignment and path planning problem for multiple CAVs on multiple lanes. A graph-based minimum clique cover method is then applied to obtain the optimal scheduling plan for the CAVs. Extensive numerical simulations for different numbers of vehicles and traffic volumes validate the effectiveness of the proposed algorithm.

Index Terms—Connected and Automated Vehicles, Lane Change Permitted Intersections, Arrival Time Scheduling, Intersection Management

I. INTRODUCTION

Intersections are the most complicated scenarios in urban traffic, where traffic jams and vehicle collisions frequently occur [1]. With the rapid development of vehicle-to-everything (V2X) technology, the central coordinator deployed at an intersection can guide connected and automated vehicles (CAVs) through it. This guarantees the high-efficiency and conflict-free cooperation of CAVs [2]. After receiving the scheduled arrival time from the coordinator, each CAV optimizes its speed trajectory in pursuit of high traffic efficiency and low fuel consumption [3], [4]. Several methods have been proposed to solve the longitudinal control problems of CAVs, including model predictive control [5], [6], fuzzy logic [7], [8], and optimal control [9], [10].

Apart from vehicle control, the CAV scheduling problem is also frequently discussed in research on intersections. CAVs approach from different directions with different destinations; thus, their trajectories inevitably intersect in the middle of the intersection. Therefore, staggering the arrival times of CAVs is an essential functionality of intersection management. Previous studies have also found that scheduling CAVs is the key

factor influencing the traffic efficiency at intersections [11], [12]. To schedule the CAVs approaching intersections, the most straightforward method is a first-in-first-out (FIFO) strategy, wherein the CAVs that enter first are scheduled to leave the intersection first [13], [14]. Similar concepts are applied in reservation-based [15], batch-based [16], and platoon-based methods [17].

Evidently, these ad hoc scheduling methods have low computational burden; however, they are less likely to obtain a high-efficiency scheduling plan. Therefore, other studies have also proposed optimization-based methods to solve the scheduling problem. Several researchers formulated this problem as a mixed integer program (MIP) problem [18], [19], [20]. Other methods, such as Monte Carlo tree search [21], dynamic programming [22], and minimum clique cover (MCC) [23], were also proposed to schedule CAVs. It is worth mentioning that because scheduling CAVs is a discrete problem rather than a continuous one, a graph-based method is another promising method to solve this problem. Apart from the depth-first spanning tree algorithm proposed in [24], a Petri net [25] and a conflict duration graph [26] have also been used in modeling the scheduling problem. Although these studies widely investigated the CAV scheduling problem, the intersection scenarios remained limited to where lane changing was prohibited.

In the aforementioned studies on intersections, the CAVs were assumed to run in their target lanes, *i.e.*, only their longitudinal control was considered. In practice, however, CAVs approach from random lanes and have different target lanes; therefore, it is necessary to extend this research to scenarios that permit lane changing. Earlier studies on this topic focused on obtaining a smooth CAV speed trajectory [27], [28]; however, traffic efficiency was not investigated completely. With regard to CAV scheduling, which we are concerned with, a few studies formulated this problem as an MIP [29] or a linear programming problem [30]. This was done by assuming that lane changing maneuvers are accomplished in a given time interval. [31] proposed a practical bi-level framework, where the high-efficiency arrival plans and collision-free path planning are solved on the upper and lower levels separately. Several other prospective studies focused on changing lane directions dynamically rather than allocating CAVs to constant directional lanes [32], [33], [34]. However, flexible lane directions are unsuitable for a mixed traffic environment, where human-driven vehicles (HDVs) and CAVs coexist.

In summary, most of the existing studies were conducted

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图 1: 首页与摘要 / Title and abstract page (p.1, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for discussing route-template/lane-choice extensions beyond the current fixed-route JSSP formulation.

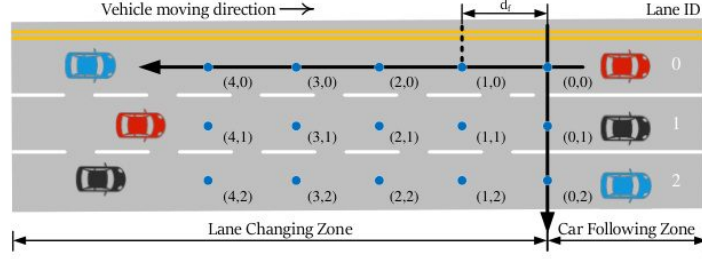


Fig. 3: Relative coordinate system (RCS). When entering the lane changing zone, the approaching CAVs are aligned to their nearest coordinates in the RCS. In this zone, the Hungarian algorithm is used to generate the target position assignment and conflict-based search (CBS) is used to find collision-free paths for the CAVs. When exiting the lane changing zone, each CAV adjusts to its target destination.

structure; thus, the CAVs fully occupy the RCS as shown in the car-following zone in Fig. 3. If there are N CAVs at primary positions (x_i^t, y_i^t) , $i \leq N$, $i \in \mathbb{N}^+$, there also exist N target positions (x_j^t, y_j^t) , $j \leq N$, $j \in \mathbb{N}^+$. Every CAV i should be assigned to a target position j .

Before assigning the target positions to the CAVs, the cost of this assignment should be defined. We use the distance from each primary position to each target position as the cost for one CAV, where the Euclidean distance is used as the evaluation index. Hence, we can obtain the cost matrix C as follows:

$$C = [c_{ij}] \in \mathbb{R}^{N \times N}, i, j \in \mathbb{N}^+, \quad (2)$$

$$c_{ij} = \sqrt{(x_k^t - x_i^t)^2 + (y_k^t - y_i^t)^2},$$

where the element in the i -th row and j -th column represents the cost to assign vehicle i to target j .

Each CAV has a preferred lane; therefore, the preference matrix $\mathcal{L}$ is obtained to define the CAV preferences.

$$\mathcal{L} = [l_{ij}] \in \mathbb{R}^{N \times N}, i, j \in \mathbb{N}^+, \quad (3)$$

$$l_{ij} = \begin{cases} 1, & \text{if vehicle } i \text{ can be assigned to target } j, \\ M, & \text{otherwise,} \end{cases}$$

where the element in the i -th row and j -th column represents whether vehicle i can be assigned to target j considering their preferred lane. M is a positive number sufficiently large to prevent vehicle i from being assigned to target j .

The assignment matrix $\mathcal{A}$, whose element in the i -th row and j -th column represents whether vehicle i is assigned to target j , is defined as follows:

$$\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N \times N}, i, j \in \mathbb{N}^+, \quad (4)$$

$$a_{ij} = \begin{cases} 1, & \text{if vehicle } i \text{ is assigned to target } j, \\ 0, & \text{otherwise.} \end{cases}$$

Subsequently, the assignment problem can be modeled as a

0-1 integer programming problem:

$$\min \sum_{i=1}^N \sum_{j=1}^N (c_{ij} \times l_{ij} \times a_{ij}), \quad (5)$$

$$\text{subject to: } \sum_{i=1}^N a_{ij} = 1,$$

$$\sum_{j=1}^N a_{ij} = 1,$$

$$i, j \in \mathbb{N}^+,$$

where N is the number of CAVs, c_{ij} is the cost matrix, l_{ij} is the preference matrix, and a_{ij} is the assignment result.

The Hungarian algorithm [41] is commonly used to solve assignment problems such as (5), and we employ this algorithm in our method to generate a feasible assignment with the lowest cost. In the following section, for each group of CAVs, we aim to obtain not only the best assignment but also several sub-optimal assignments. For simplicity, we use A_k to represent the k -th optimal assignment and C_k to denote the corresponding cost C .

C. Conflict Types in Lane Changing Behavior

After obtaining the assignment, we need to develop path planning for the CAVs, which should not have conflicts in their trajectories. First, the collision types in the lane changing behavior of CAVs should be clarified. Notably, we regulate the CAVs to move only along orthogonal directions in the RCS, *i.e.*, movement along diagonal directions in the RCS is prohibited. Movement along diagonal directions in the RCS produces large acceleration and deceleration in lane change behavior, which is hazardous and also increases the computational burden in multi-vehicle path planning. Hence, we constrain the movement of the CAVs to orthogonal directions in the RCS for simplicity.

图 2: 建模或问题设定页 / Modeling or problem-setup page (p.4, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for discussing route-template/lane-choice extensions beyond the current fixed-route JSSP formulation.

in scenarios where lane changing was prohibited. Therefore, only CAV scheduling or vehicle longitudinal control were studied. In this study, we focus on a scenario where lane changing is permitted. CAVs are required to change to their target lanes on approaching the intersection. Most of the aforementioned studies circumvent this problem by assuming that the CAVs are running in their target lanes [11], [13], [15], [16], [24]. A few studies considered lane changing behavior in their scheduling algorithm [29], [30], [31]; however, the optimality of the scheduling was not extensively discussed. Moreover, although CAV cooperation control at intersections has undeniable advantages in terms of collision avoidance, constant traffic signal phase and timing (SPAT) control also has advantages in terms of vehicle evacuation, especially for large traffic volumes. In this respect, the comparison between CAV cooperation control and constant traffic SPAT remains deficient.

Thus, the main contributions of this study are as follows. We propose a two-stage cooperation framework to decouple the longitudinal and lateral control of CAVs. In the first stage, a formation control method is used to guide the CAVs into their target lanes. An iterative framework is developed to solve the multi-vehicle target assignment and path planning problem, preventing the deadlock problem in the single vehicle lane changing algorithm. In the second stage, the CAV arrival time is optimized to increase the traffic efficiency. Specifically, a heuristic graph-based solution is proposed to solve the scheduling problem with low computational burden. As opposed to specially considering lane changing during algorithm design [29], [30], [31], the decoupling framework has an improved generalization ability in merging with other scheduling methods. In simulations, we compare the proposed algorithm with constant traffic SPAT, and our results verify the effectiveness of the proposed algorithm.

The remainder of this paper is organized as follows. Section II describes the scenario and introduces the two-stage cooperation framework. Section III explains the multi-vehicle target assignment and path planning method. Section IV presents the graph-based CAV scheduling method. The simulation results are discussed in Section V, and Section VI concludes the paper.

II. TWO-STAGE COOPERATION FRAMEWORK

In [23], we introduced a graph-based minimum clique cover (MCC) method applied to an intersection with complex vehicle conflict relationships, as shown in Fig. 1. In this previous study, changing lanes was prohibited to guarantee vehicle safety, *i.e.*, the CAVs are assumed to be in their target lane from the beginning. In this study, the same intersection is used as the traffic scenario; however, lane change is permitted. The CAVs have different destinations; therefore, it is apparent that a scenario where lane changing is permitted is more realistic. Moreover, two assumptions remain to be clarified.

Assumption 1: CAVs transmit their velocities and positions to the central cloud coordinator through ideal wireless communication, *e.g.*, V2I communication [35], where communication delay and packet loss do not occur.

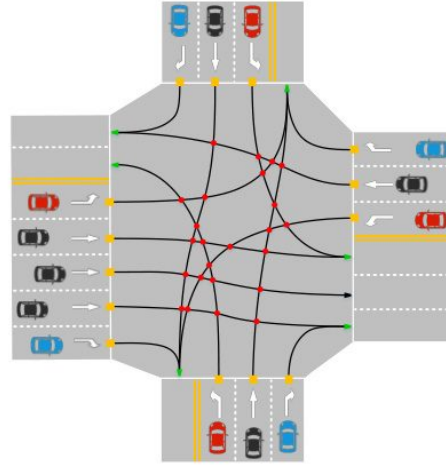


Fig. 1: Illustration of the traffic scenario. Vehicles are colored in red, black, and blue to represent turning left, going straight, and turning right, respectively. Red circles, orange squares, and green arrows represent different types of potential collision points.

Assumption 2: The CAVs are capable of fully autonomous driving, implying that they are assumed to have perfect steering performance in their lane changing and turning behavior.

Considering the limit range of V2X communication, previous studies usually set a certain range of the *control zone* to design cooperation algorithms [2], [3], [23], [24]. In our study, we aim to decouple the longitudinal and lateral control problem in CAVs; therefore, the *control zone* is divided into a *lane changing zone* and a *car-following zone*, as shown in Fig. 2, whose lengths are denoted as L_{ctrl} , L_{LCZ} , and L_{CFZ} , respectively.

Most of the existing studies only considered the trajectory planning of CAVs, *i.e.*, their longitudinal control, implying that their target lane selection and lane changing behavior was neglected in intersection management. In this study, we propose a two-stage cooperation framework to decouple the longitudinal and lateral control of CAVs, as shown in Fig. 2. The lane changing behavior is strongly related to the surrounding CAVs, and an *edge coordinator* is deployed at the lane changing zone to coordinate the movement of the CAVs to their target lanes. When entering the lane changing zone, task assignment is first accomplished to allocate the preferred lanes to the CAVs. We then develop multi-vehicle path planning to guarantee a collision-free scenario during lane changing. Therefore, when exiting the lane changing zone, all CAVs are aligned to their target positions. As the CAVs enter the car-following zone, the *central coordinator* collects the information from all of them at the intersection and generates

图 3: 核心方法或算法页 / Core method or algorithm page (p.2, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for discussing route-template/lane-choice extensions beyond the current fixed-route JSSP formulation.

Types	Parameter	Symbol	Value
Simulation	Simulation step	-	0.1s
	Lane changing zone length	L_{LCZ}	500m
	Car-following zone length	L_{CFZ}	500m
RCS	Group size	-	3
	One-step time	-	4s
	Lane changing time	-	3s
Controller	Feedback gain of distance error	k_p	0.1
	Feedback gain of velocity error	k_v	0.3
	Steady platoon velocity	v_p	10m/s
	Steady car-following distance	d_f	30m
Constraints	Maximum acceleration	a_{max}	5m/s ²
	Minimum acceleration	a_{min}	-6m/s ²
	Maximum velocity	v_{max}	15m/s
	Minimum velocity	v_{min}	0m/s

TABLE II: Parameters

A linear feedback controller is designed as follows:

$$u_i = - \sum_{j \in \mathbb{I}_i} k_p \delta_p^{(i,j)} - \sum_{j \in \mathbb{I}_i} k_v \delta_v^{(i,j)}, \quad (16)$$

where k_p and k_v are the feedback gains of the distance and velocity errors of CAV i , respectively. The same gains are set for all the CAVs because we consider a homogeneous scenario.

As mentioned earlier, we consider second-order vehicle dynamics as shown in (9). We define the car-following errors as the new vehicle state.

$$\bar{x}_i = \begin{bmatrix} \bar{x}_{i,1} \\ \bar{x}_{i,2} \end{bmatrix} = \begin{bmatrix} p_0 - p_i - D_{des}(d_0 - d_i) \\ v_i - v_p \end{bmatrix}, i \in \mathbb{N}^+. \quad (17)$$

The vehicle input remains the same, *i.e.*, $\bar{u}_i = u_i$.

Therefore, the car-following vehicle dynamic model is

$$\dot{\bar{x}}_i = \mathbf{A}\bar{x}_i + \mathbf{B}\bar{u}_i, i \in \mathbb{N}^+. \quad (18)$$

The linear feedback controller is simplified to

$$\bar{u}_i = -k_p \sum_j (l_{ij} + q_{ij}) \bar{x}_{j,1} - k_v \sum_j (l_{ij} + q_{ij}) \bar{x}_{j,2}, j \in \mathbb{I}_i. \quad (19)$$

Defining $\mathbf{k} = [k_p, k_v]^T$, we have

$$\bar{u}_i = - \sum_j (l_{ij} + q_{ij}) \mathbf{k}^T \bar{x}_j, i \in \mathbb{N}^+. \quad (20)$$

V. SIMULATIONS

A. Simulation Environment

The traffic simulation was conducted in SUMO, which is widely used in traffic research[52]. The simulation was run on an Intel Core i7-7700 @3.6 GHz processor. The intersection scenario and lane direction settings are the same as shown in Fig. 1, and the overall length of the lane changing and car-following zones is 1000 m. The CAV arrival is assumed to be a Poisson distributed flow, given by

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, k = 0, 1, \dots, \quad (21)$$

where X represents the arrival of the vehicle at the control zone. λ is the expected value as well as the variance of

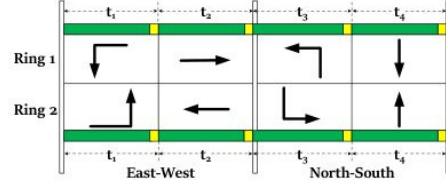


Fig. 9: Dual ring control in constant traffic SPAT. The green line indicates that the corresponding traffic light is set to green, and the yellow square denotes the clearance time.

the Poisson distribution. Other key simulation parameters are listed in Table II.

B. Evaluation Indexes

As mentioned earlier, we aim to propose a cooperation method to improve both traffic safety and efficiency, *i.e.*, obtain the collision-free optimal CAV passing order. We assign t_i^{in} as the time step when vehicle i enters the lane changing zone and t_i^{out} as the time step when it arrives at the intersection. Several evaluation indexes have been proposed to measure the scheduling performance. In this study, we chose the evacuation time as the primary optimization target, as shown in Definition 5. The evacuation time is related to the spanning tree depth, which is described in Section IV.

Definition 5 (Evacuation Time): The evacuation time of N CAVs is defined as the time when the last CAV reaches the stop line; it is expressed as

$$t_{\text{evc}} = \max t_i^{\text{out}}, i \leq N, i \in \mathbb{N}^+. \quad (22)$$

Considering N incoming CAVs, t_{evc} represents the arrival time of the last CAV at the stop line. For N CAVs, smaller evacuation times indicate that these CAVs pass through the intersection in a shorter time. Thus, it demonstrates the overall traffic efficiency performance, *i.e.*, the overall benefits to the CAVs.

In addition, the vehicle travels through the control zone in $t_i^{\text{out}} - t_i^{\text{in}}$ time, whereas it travels through it under the free driving condition in $L_{\text{ctrl}}/v_{\text{max}}$ time. Accordingly, the ATTD is selected to measure the average traffic efficiency of the vehicles, as described in Definition 6.

Definition 6 (Average Travel Time Delay): The ATTD is designed to evaluate the average traffic efficiency of N CAVs. It is expressed as

$$t_{\text{ATTD}} = \frac{1}{N} \sum_{i=1}^N \left(t_i^{\text{out}} - t_i^{\text{in}} - \frac{L_{\text{ctrl}}}{v_{\text{max}}} \right), \quad (23)$$

where L_{ctrl} is the length of the control zone and t_{ATTD} represents the average travel delay of the CAVs. The travel time of every CAV is considered in t_{ATTD} ; therefore, it denotes the individual benefits for N CAVs, which is the secondary optimization target of traffic efficiency.

图 4: 实验或结果页 / Experiments or results page (p.10, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for discussing route-template/lane-choice extensions beyond the current fixed-route JSSP formulation.

The simulation result is shown in Fig. 10. The bar plot shows the average value of the ten sets of simulations. The standard deviations of each set of results are also provided. Generally, both the evacuation time and the ATTD increase with the number of vehicles because of vehicle queuing. It is apparent that the MCC in the lane changing zone significantly improves the traffic efficiency. The evacuation time is the primary optimization target in the MCC scheduling; therefore, the MCC algorithm reduces the evacuation time by 15.8% to 67.9% for 20 to 100 vehicles.

Regarding the ATTD perspective, it is evident that the FCLC negatively influences the ATTD. This is because when SumoLC is applied, CAVs are set to the maximum velocity $v_{\max} = 15 \text{ m/s}$ if there are no other vehicles around. However, the FCLC aims to develop a steady parallel formation with a platoon velocity $v_p = 10 \text{ m/s}$. Therefore, the total evacuation time is unaffected when vehicle number is less than 80 because the intersection is not oversaturated; however, the ATTD is higher because of the delay in the lane changing zone. On the other side, the formation control in FCLC helps to generate a collision-free and feasible path planning set to MCC. Moreover, the steady parallel formation is convenient for MCC to adjust the car following distance to desired virtual platoon positions. MCC algorithm reduces 2.2% to 32.6% of the ATTD. Moreover, as the vehicle number increases, it has less improvement on ATTD since ATTD is not the priority concern in optimization.

E. Simulation Results for Different Traffic Volumes

The second simulation was conducted for different traffic volumes. Similar to the previous simulation, ten randomly generated repetitions of the simulation were conducted for each traffic volume. The number of vehicles was set to 50, and the traffic volume varied from 500 veh/h to 2500 veh/h.

Generally, as the number of vehicles is constant, the evacuation time is decreased as the traffic volume increases. The MCC exhibits noticeable improvement in the traffic efficiency, which reduces the evacuation time by 18.8% to 33.2% and ATTD by 1.9% to 14.5%. The comparison between the SumoLC and FCLC results agrees with the analysis of the simulation results for different numbers of vehicles. FCLC delays the travel time in lane changing zone because of velocity settings. FCLC+MCC provides a feasible solution with noticeable improvement in the traffic efficiency.

VI. CONCLUSIONS

In this paper, a two-stage cooperation framework is proposed to improve vehicle safety and traffic efficiency at intersections where lane changing is permitted. In the first stage, we design an iterative method to solve the multi-vehicle target assignment and path planning problem. In contrast to existing single-vehicle lane changing algorithms, the deadlock problem is solved in our method and a feasible collision-free path planning is ensured. In the second stage, a graph-based method is proposed to schedule the CAV arrival time. A heuristic algorithm is established to solve the problem with

a low computational burden. Traffic simulations verified the effectiveness of the proposed algorithm.

Future research directions include studying the relationship of the two decoupled stages further. In this study, the optimal arrival plan is calculated when the CAV arrives at the car-following zone. However, it can also be calculated when the CAV enters the lane changing zone. Thus, in the lane changing zone, a CAV can drive to its target position in virtual platoon. Another interesting topic involves considering HDVs in this scenario because lane changing would be permitted. There has been little research on intersections with mixed traffic where lane changing is permitted.

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图 5: 结论或局限页 / Conclusion or limitations page (p.13, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for discussing route-template/lane-choice extensions beyond the current fixed-route JSSP formulation.

3 章节精读与逐段导读 / Section-by-Section Deep Reading & Walkthrough

p.1 I. INTRODUCTION

章节目标 / Learning goal: 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 Lane changing + arrival scheduling：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.2 II. TWO-STAGE COOPERATION FRAMEWORK

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 框架先求多车目标分配与路径规划，再基于最终路径构建冲突图并求到达时间。这样横向变道改变冲突关系，纵向调度再处理被改变后的路口资源竞争。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.3 III. MULTI-VEHICLE TARGET ASSIGNMENT AND PATH

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.6 1. However, we

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.7 IV. CAV SCHEDULING AT UNSIGNALIZED INTERSECTIONS

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.9 4. This also indicates that the theoretical evacuation times

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

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审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.9 6. We have proved that this can be solved by exchanging the

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.10 V. SIMULATIONS

章节目标 / Learning goal: 验证方法是否在公平基准下改善效率、安全、能耗或计算时间。

Check whether the claimed improvements are supported by fair baselines and meaningful metrics.

关键论点 / Key argument: 实验应与论文声称的贡献闭环: Numerical simulations for different vehicle counts and traffic volumes.

技术焦点 / Technical focus: 重点追踪指标: 平均延误、总行程时间、完成变道比例、冲突避免成功率和交通量敏感性。同时检查仿真平台、交通需求和随机性设置。

审稿追问 / Reviewer question: 实验是否只展示平均收益，还是覆盖了高密度、异常扰动和失败案例？

3.1 逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段 先把作者的痛点翻译成一句博士问题: 当车辆可以变道或选择目标车道时，如何联合决定路径/车道与无冲突到达时间？读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段 把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：车辆可执行规划出的变道动作，周围交通允许形成编队。；车道级地图和目标车道需求已知。；变道安全约束可以被简化成可规划的路径/队形约束。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段 围绕核心公式“联合车道分配与到达时间优化 / Joint lane assignment and arrival scheduling”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段 把算法读成一条数据流：输入交通状态，执行 Two-stage framework decoupling lateral lane assignment/path planning from graph-based arrival scheduling with minimum clique cover.，输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段 不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Numerical simulations for different vehicle counts and traffic volumes.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段 最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Useful for discussing route-template/lane-choice extensions beyond the current fixed-route JSSP formulation.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

4 技术路线图与贡献量评估 / Technical Route & Contribution Matrix

Step	Stage / 阶段	Content / 内容
1	输入 / Input	车辆状态、路径/车道、交通需求、信号或协同信息。
2	建模 / Modeling	联合车道分配与到达时间优化 / Joint lane assignment and arrival scheduling
3	核心求解 / Core solver	Two-stage framework decoupling lateral lane assignment/path planning from graph-based arrival scheduling with minimum clique cover.
4	安全接口 / Safety interface	Conflict-free arrival scheduling and formation-control assumptions prevent incompatible movements.

5	平滑与效率 / Smoothing and efficiency	Improves traffic efficiency by choosing target lanes and arrival schedules that avoid unnecessary queuing.
6	验证 / Validation	Numerical simulations for different vehicle counts and traffic volumes.

Dimension / 维度	Score	Comment / 评语
理论贡献 / Theoretical contribution	3/5	联合车道分配与到达时间优化 / Joint lane assignment and arrival scheduling
算法贡献 / Algorithmic contribution	4/5	Two-stage framework decoupling lateral lane assignment/path planning from graph-based arrival scheduling with minimum clique cover.
实验贡献 / Experimental contribution	3/5	Numerical simulations for different vehicle counts and traffic volumes.
工程贡献 / Engineering contribution	3/5	Conflict-free arrival scheduling and formation-control assumptions prevent incompatible movements.
博士课题价值 / PhD relevance	4/5	Useful for discussing route-template/lane-choice extensions beyond the current fixed-route JSSP formulation.

5 理论方法论深度解构 / Methodology & Theoretical Framework

5.1 问题数学建模 / Mathematical Formulation

联合车道分配与到达时间优化 / Joint lane assignment and arrival scheduling

$$\min_{l_i, t_i} \sum_{i \in V} \left(\alpha d_i(l_i) + \beta |t_i - t_i^{des}| \right) \quad \text{s.t. } (i, j) \in E(l) \Rightarrow |t_i - t_j| \geq \tau_{ij}$$

Symbol / 符号	含义	Meaning	Role / 建模作用
l_i	目标车道	target lane	横向规划决策变量
t_i	车辆到达冲突区时间	arrival time at conflict zone	纵向调度决策变量
$d_i(l_i)$	变道代价	lane-changing cost	衡量横向操作复杂度
$E(l)$	由车道选择诱导的冲突边集	lane-induced conflict edges	说明横向选择会改变调度约束
τ_{ij}	安全时间间隔	safety headway	保证冲突车辆时间分离

5.2 核心算法架构 / Core Algorithm Layout

框架先求多车目标分配与路径规划，再基于最终路径构建冲突图并求到达时间。这样横向变道改变冲突关系，纵向调度再处理被改变后的路口资源竞争。

The framework first assigns target lanes/paths, then constructs a conflict graph for those paths and computes a conflict-free arrival schedule.

5.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

分阶段设计提高实时性，但牺牲联合最优性；如果第一阶段车道选择失误，第二阶段只能在较差冲突结构上补救。

The two-stage decomposition is tractable but not globally optimal; poor lane assignment can constrain the downstream scheduler.

5.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 车辆可执行规划出的变道动作，周围交通允许形成编队。
- 车道级地图和目标车道需求已知。
- 变道安全约束可以被简化成可规划的路径/队形约束。

6 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境：固定车道策略、无变道调度、FCFS 和基于图的到达调度。

Validation / 验证：Numerical simulations for different vehicle counts and traffic volumes.

Metrics / 指标：平均延误、总行程时间、完成变道比例、冲突避免成功率和交通量敏感性。

Ablation / 消融：最关键消融是去掉 lane-changing stage，对比是否真的改善调度；还应测试高密度下变道失败的退化策略。

Reviewer reading / 审稿式阅读提醒：阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

6.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 44 figure references and 6 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。

冲突关系图 / 通行顺序图	conflict-relation or passing-order graph	把节点、边类型和先后约束翻译成 JSSP 的作业、机器和析取约束。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

7 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

7.1 论文局限性 / Limitations & Weaknesses

- 文本与同作者早期图调度工作高度相关，独立贡献需谨慎引用。
- 横向控制细节和人类驾驶干扰不足。
- 两阶段解耦可能错过更优的路径-时间联合解。

7.2 博士课题拓展点 / Research Extensions for PhD

- 用双层优化 (bi-level optimization) 同时学习车道选择和 JSSP 排序。
- 为变道失败设计可恢复调度 (recoverable scheduling)，让路口管理器在线回滚目标车道。
- 把车道选择看成 route-template selection，扩展当前固定路线假设。

7.3 如何服务当前课题 / Use in This Project

Useful for discussing route-template/lane-choice extensions beyond the current fixed-route JSSP formulation.

8 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter: 可变道车辆比例 / Lane-change-enabled ratio, range $[0, 1]$.

比例增加会提高调度自由度，但过高时变道冲突和局部扰动也会增加。

More lane-change freedom improves scheduling flexibility but can add lateral conflicts.

8.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar: 题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator，右侧为五大精读模块。
- 公式卡片使用 `<pre>` 或 LaTeX block，紧跟符号表。
- 审稿人批注用 reviewer-note block，博士拓展用 research-idea list。
- 交互参数用 `input[type=range]` + canvas 曲线，打印 PDF 时隐藏交互控件但保留解释。

9 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

Page	Extracted heading
p.1	I. INTRODUCTION
p.2	II. TWO-STAGE COOPERATION FRAMEWORK
p.10	V. SIMULATIONS

p.1 I. INTRODUCTION 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。/ Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.2 II. TWO-STAGE COOPERATION FRAMEWORK 给出算法流程和求解机制，应重点追踪输入、输出和安全接口。/ Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.3 III. MULTIVEHICLE TARGET ASSIGNMENT AND PATH 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.6 1. However, we 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.7 IV. CAV SCHEDULING AT UNSIGNALIZED INTERSECTIONS 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.9 4. This also indicates that the theoretical evacuation times 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.9 6. We have proved that this can be solved by exchanging the 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.10 V. SIMULATIONS 检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。/ Tests efficiency, safety, or energy claims; inspect whether baselines are fair.